

DevOps Coding Questions (Bash, YAML, Docker, Terraform)

1. Loop through files in a directory and print line count

```
for file in *.log; do
    echo "$file has $(wc -l < "$file") lines"
done
```

2. Extract lines with 'error' from log (case-insensitive)

```
grep -i "error" app.log
```

3. Delete files older than 7 days

```
find /var/log/myapp -type f -mtime +7 -exec rm -f {} \;
```

4. List only directories from /etc

```
ls -l /etc | grep "^d"
```

5. Create a cron job to restart Nginx every night at 2 AM

```
0 2 * * * /bin/systemctl restart nginx
```

6. Script to monitor disk usage and alert if >80%

```
#!/bin/bash

threshold=80

usage=$(df / | grep / | awk '{ print $5 }' | sed 's/%//')

if [ "$usage" -gt "$threshold" ]; then
    echo "Disk usage critical: $usage%" | mail -s "Disk Alert" admin@example.com
fi
```

7. Basic Dockerfile for Nginx

```
FROM nginx:alpine

COPY ./index.html /usr/share/nginx/html/
```

8. GitLab CI/CD YAML

```
stages:
```

- build
- deploy

build-job:

```
stage: build

script:
  - echo "Building app..."
  - ./build.sh
```

deploy-job:

```
stage: deploy

script:
  - echo "Deploying..."
  - ./deploy.sh
```

9. Kubernetes Pod Definition (YAML)

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:alpine
      ports:
        - containerPort: 80
```

10. Terraform Variable Example

```
variable "region" {
  default = "us-central1"
}

resource "google_compute_instance" "vm" {
  name          = "my-vm"
```

```
machine_type = "e2-micro"  
zone         = "${var.region}-a"  
}
```